

Algorithm Sequencing in Deep Reinforcement Learning

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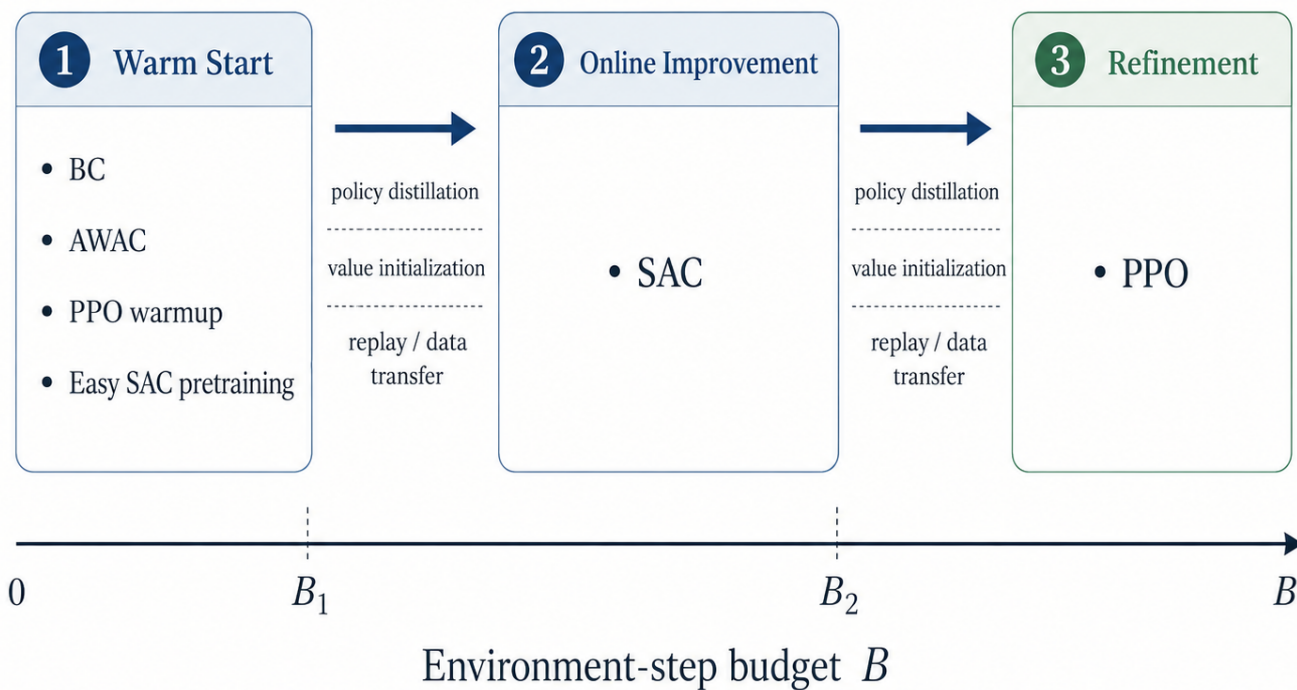
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Question and Hypothesis

- RL can run as one fixed algorithm, but training has phases: initialization, exploration, improvement, and refinement.
- Can **algorithm sequencing** improve final return, sample efficiency, or stability under matched online budgets?
- Hypothesis: sequencing helps when the first phase gives the next phase a useful policy, state distribution, or representation while leaving enough budget to improve.
- The contribution is not “SAC beats PPO”: **phase compatibility matters and can be understood.**

Phase-Based Scheduler



Experiment Map

Family	Question	Evidence
Single algorithms	How strong are fixed baselines?	PPO, SAC; 5 seeds; Hopper/Walker.
Online handoffs	Does switch direction/timing matter?	SAC→PPO, PPO→SAC; 25/50/75% and plateau switches.
Transfer mechanism	Does value warmup fix SAC→PPO?	Random, self-warmup, source-aligned PPO values.
Offline starts	Can BC/AWAC initialize online RL?	BC→SAC, AWAC→SAC, interleaved BC anchoring, PPO targets.
Curriculum starts	Does an easier first phase help?	Easy SAC→SAC on Hopper; “easy” ignores early termination during pretraining.

What We Measure

- Final return:** end-of-budget policy quality.
- Normalized AUC:** sample efficiency over the full learning curve.
- Stability:** seed variance, collapse count, and worst-seed return.
- Budget accounting:** online environment steps are matched; offline data and updates are reported separately.

Design Principle

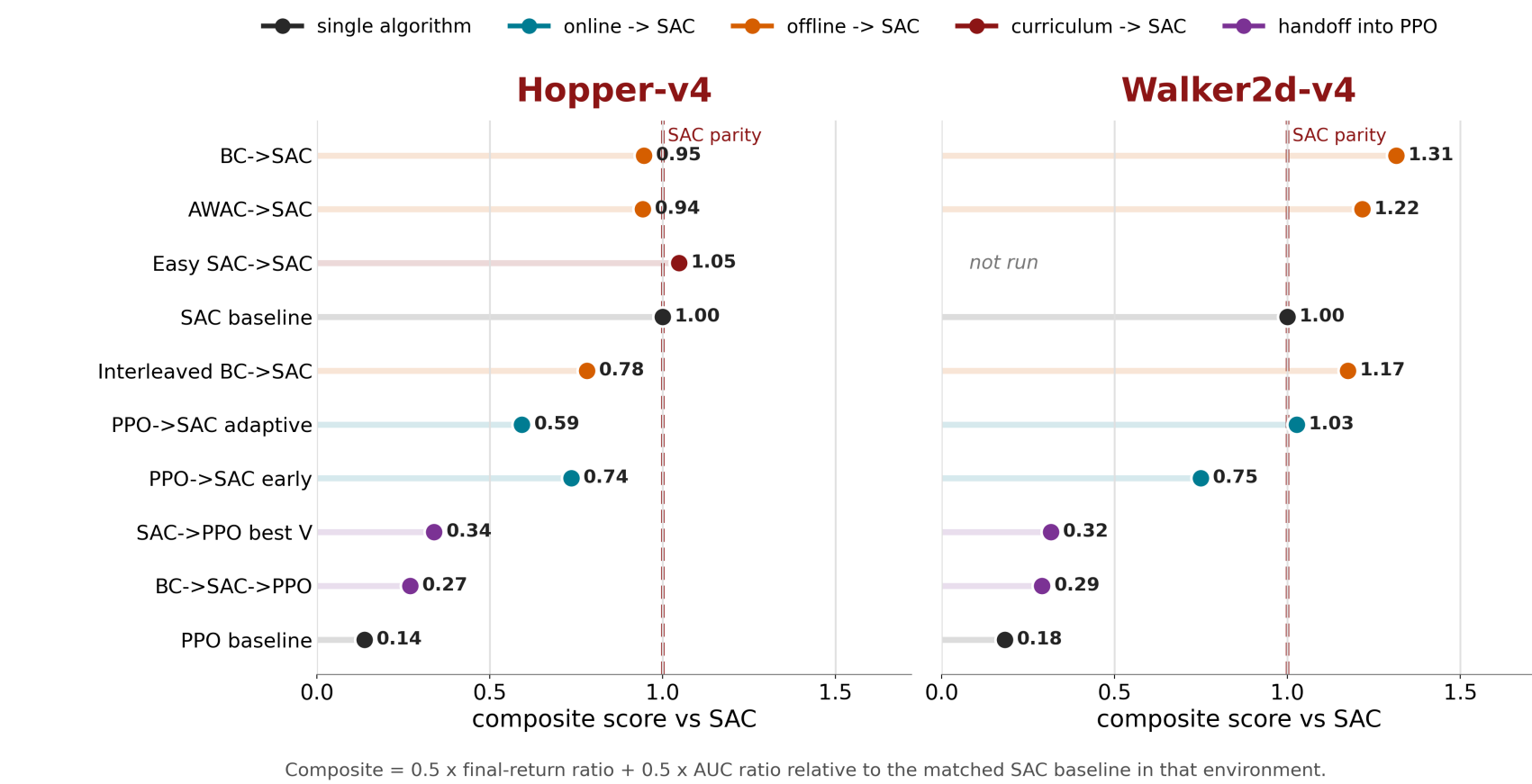
- Warm starts should produce behavior the online learner can preserve and improve.
- Exploration phases should hand off early enough for the target algorithm to use most of the remaining budget.
- Refinement phases need transfer mechanisms that prevent policy degradation; value initialization alone is weak evidence for this.

Motivation: Three-Stage Policy Training



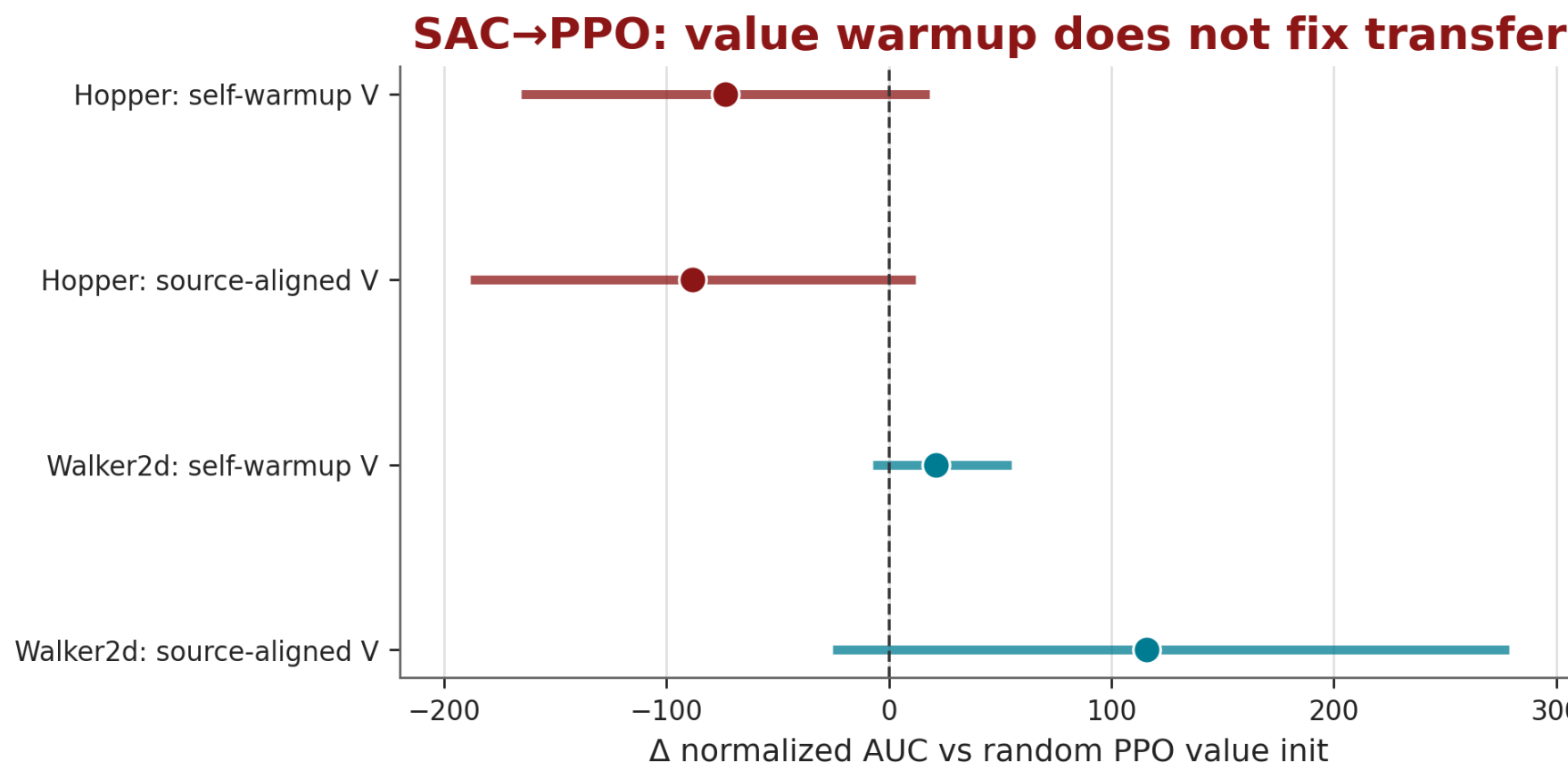
Main Result: Compatible Schedules End in SAC

Synthesis: schedules that end in SAC dominate the phase-allocation frontier



- SAC remains the strongest fixed baseline, so a useful scheduler must justify its phase allocation against SAC.
- The best non-baseline schedules hand into SAC: PPO→SAC, BC→SAC, AWAC→SAC, interleaved BC→SAC, and Easy SAC→SAC.
- Handoffs into PPO preserve neither final return nor AUC in our current runs, even when they receive a pretrained policy.

Transfer Out of SAC Is Fragile



95% paired CIs cross zero: critic/value transfer is not the primary bottleneck.

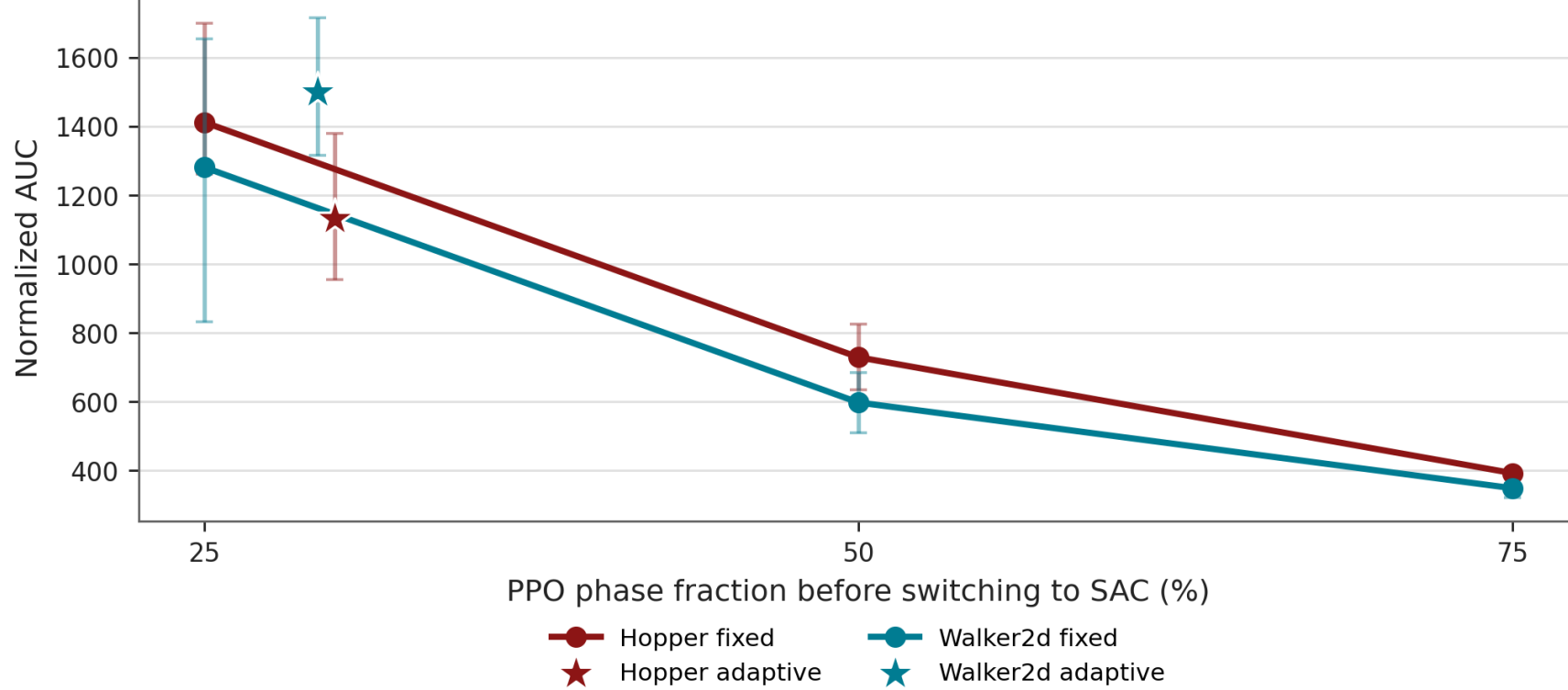
- SAC→PPO did not become reliable after PPO value warmup compared to SAC only
- Mismatch between source trajectory and target learner preventing all handoffs from learning
- Interpretation:** PPO refinement needs a stronger policy-retention or state-distribution transfer mechanism before it can be a dependable final phase.

Curriculum / Transfer Learning

Easy Hopper: We modify Hopper-v4 only during pretraining by ignoring early failure terminations. If the hopper falls, the episode continues instead of resetting, giving SAC longer trajectories before transferring the learned policy to normal Hopper-v4.

PPO→SAC Needs Early or Adaptive Switching

PPO→SAC: switch early or adaptively



Stars show plateau-triggered adaptive switches; both fire near the strong early-switch regime.

- Fixed 25% PPO phases beat late 75% switches; SAC needs enough remaining budget to improve the distilled policy.
- Plateau switches fire near the strong early-switch regime and avoid wasting budget on an exhausted first phase.

Offline and Curriculum Starts Work Best Into SAC

Offline/curriculum warm starts: SAC is the best target

BC→SAC	3047	1898	3965	2351
AWAC→SAC	2609	2225	3545	2236
Interleaved BC→SAC	1995	1975	2751	2493
Easy SAC→SAC	3333	2134	—	—
BC→PPO	503	553	626	544
AWAC→PPO	518	527	794	459
BC→SAC→PPO	472	852	372	770
	Hopper final	Hopper AUC	Walker final	Walker AUC

Cell color is normalized within each metric column. Easy SAC→SAC was run on Hopper only.

- BC/AWAC starts transfer effectively into SAC; PPO targets remain weak after the same starts.
- Interleaved BC anchoring improves Walker AUC, while Easy SAC→SAC shows Hopper curriculum pretraining can improve final return after real-environment fine-tuning.

Takeaways and Next Steps

- Sequencing is useful when it creates a better initialization for SAC, not as a universal replacement for SAC.
- Current best scheduler family: warm start or short PPO phase → SAC, with adaptive switching to reduce timing sensitivity.
- Open problem: make PPO a real stabilization phase by transferring more than a policy snapshot or value function.